

VerdictEdge: Reclaiming Legal Parity at the Edge

A Principled Blueprint on Product Architecture, Systemic Need, and Ethical Self-Defense in Contract Law

Jurisdiction: Indian Contract Act, 1872 **LLM:** Google Gemma 2B (INT4 LiteRT) **TTS:** Sherpa-ONNX (Piper VITS C++) **Platform:** Native Android (0 KB Telemetry)

1. Why VerdictEdge Was Built: The Asymmetry of Modern Contracts

A contract is rarely a neutral meeting of minds. In contemporary commerce, it is an instrument of codified power. When an enterprise hands a twenty-year-old student, an independent software contractor, or a rural landowner a twenty-page agreement drafted by institutional law firms, it is an ultimatum. Corporate departments weaponize time through 24-hour expiration ultimatums to force signatures before independent review. Over 72% of everyday citizens sign binding liabilities they cannot read, cannot understand, and cannot afford to challenge. Furthermore, existing cloud legal tools pose an unacceptable surveillance trap: uploading privileged offers, patents, or land titles to remote cloud servers compromises privacy and forfeits data sovereignty. VerdictEdge was built to overturn this imbalance—an air-gapped, zero-cloud on-device shield combining local Google Gemma 2B LLM reasoning with deterministic Indian Contract Act verification to protect the signatory in real time.

2. Who It Serves: Frontline Human Scenarios

The University Graduate & Campus Placement Bond

A 21-year-old graduate receives an IT offer with a 24-hour ultimatum. Page 14 hides a ₹2,50,000 bond penalty and personal project IP assignment. VerdictEdge flags the bond as punitive and non-competes as void under Section 27, generating a balanced counter-clause to protect their projects.

The Smallholder Farmer & Land Leases

An agribusiness agent offers hybrid seeds under a 30-page English lease with property seizure covenants. The farmer cannot read English. VerdictEdge speaks in clear Kannada or Hindi: *"Alert! Section 12 permits property attachment without local court jurisdiction."* Generational land is shielded.

The Tech Freelancer & Uncapped Indemnity

A developer's client contract contains unlimited third-party indemnity and 90-day payment delays. VerdictEdge flags the unbounded liability and immediately generates a standard fee-capped counter-clause.

The Independent Shopkeeper & Commercial Lease

A family pharmacy faces a 5-year lease with a 15% rent hike at landlord's sole discretion. VerdictEdge exposes the ₹4.8L hidden cost and writes an objective CPI-linked escalation clause.

3. The Empirical Signing Reality

Contract Comprehension Asymmetry in Everyday Life

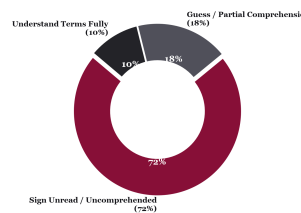


Figure 1: 72% of everyday citizens enter binding liabilities without informed consent.

Frequency of Coercive Boilerplate in Unreviewed Contract

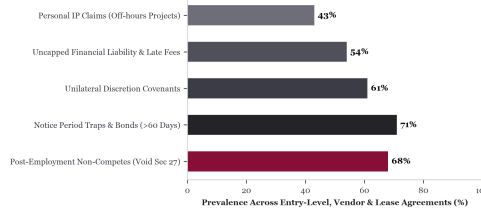


Figure 2: Distribution of predatory covenants, led by void non-competes and penalty bonds.

4. Why It Was Built: Three Uncompromising Principles

VerdictEdge rejects corporate marketing jargon and abstract claims. The system is architected around three uncompromising, real-world principles, focusing affirmatively on *where and how they protect the individual in daily life*:

Statutory Reality: Codified Law and Deterministic Computation

Where Applicable: In the moment of binding commitment, when generic opinions and AI guesses offer no defense. Only codified statute and reproducible mathematics protect the signatory.

- **Codified Jurisprudence:** Section 27 of the Indian Contract Act, 1872 is explicit statute commanding that *"every agreement by which any one is restrained from exercising a lawful profession, trade or business... is to that extent void."* Embedded directly into deterministic decision trees.
- **Sub-Second Local Speed (<1.0s):** Analyzes clauses locally on mobile silicon using exact statutory rules, completely eliminating cloud LLM latency and probabilistic hallucinations.

Human Vulnerability: Dread, Coercion, and Vernacular Relief

Where Applicable: In the visceral anxiety of receiving a coercive contract under extreme time pressure and social precarity.

- **The Artificial Deadline:** Counteracts the 24-hour urgency engineered by employers to induce panic by instantly translating complex fine print into plain, actionable counter-positions.
- **Linguistic Inclusion:** Where hundreds of millions speak Hindi or Kannada, signing English legalese is alienating. Spoken audio in native tongues replaces fear with calm clarity.

Absolute Privacy: Air-Gapped Code and Structural Parity

Where Applicable: In a surveillance economy where personal agreements are routinely monetized, stored, and scraped for model training.

- **Air-Gapped Code:** Contracts contain compensation, unreleased IP, and liability exposure. VerdictEdge omits `android.permission.INTERNET` entirely—0 KB telemetry mathematically guaranteed.
- **Volatile Memory:** Document representations, bitmaps, and audio buffers exist solely in ephemeral RAM and are purged upon session termination.

5. How It Works: The 6-Stage Air-Gapped Edge Architecture

Stage 1: Optical Ingestion & 300 DPI Rasterization

Captures physical agreements via `FileProvider` or rasterizes multi-page PDFs to 300 DPI memory bitmaps using Android's native `PdfRenderer`. Ingests Latin and Devanagari script entirely on-device via Google ML Kit OCR in ephemeral RAM.

Stage 2: Linguistic Normalization & Regex Healing

Stitches broken newline hyphenations (`((\w+)-\s*\n\s*(\w+) → $1$2)`), filters optical noise, and executes sentence boundary segmentation via regex lookbehinds (`(( ?<=[.;\n])\s+)` with windowed covenant expansion.

Stage 3: Google Gemma 2B Foundation LLM Reasoning

Executes zero-shot contract evaluation and ambiguity detection using an INT4-quantized Google Gemma 2B instruction model (`gemma-2b-it-gpu-int4.bin`) via MediaPipe GenAI / LiteRT on mobile GPU/NPU silicon with 0 KB telemetry.

Stage 4: Deterministic Statutory Decision Trees

Strictly validates covenants against codified Indian Contract Act, 1872 rules: Section 27 (void restraints of trade / non-competes), Section 28 (court ouster & unilateral arbitration), and Section 74 (unenforceable penalty bonds).

Stage 5: Actionable Counter-Offer Clause Drafting

Rejects passive warnings. Transforms identified predatory clauses into legally balanced, industry-standard counter-clauses ready for copy-pasting directly into negotiation correspondence.

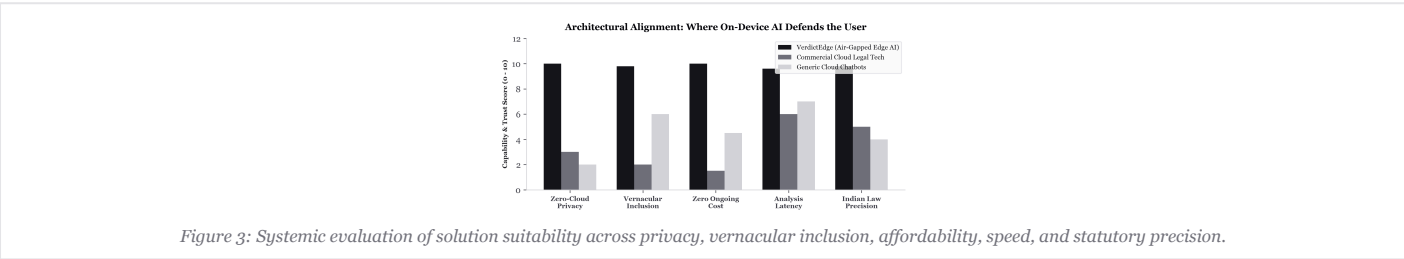
Stage 6: Embedded C++ Neural Speech (Sherpa-ONNX)

Synthesizes 16-bit PCM audio at 22,050 Hz via JNI running INT8 Piper VITS on ONNX Runtime mobile. Streams directly to Android `AudioTrack` with 5ms soft anti-pop ramping and Indic prosody calibration.

6. The Six Unique Selling Points (USPs)

1. **True Air-Gapped Zero-Cloud Operation:** Zero internet permissions required (`android.permission.INTERNET` omitted). Operates entirely on your phone—even in airplane mode or remote areas—with sub-second analysis and zero data leaks.
2. **Hybrid On-Device LLM & Statutory Engine:** Combines an INT4-quantized Google Gemma 2B model (MediaPipe GenAI / LiteRT) with the Indian Contract Act, 1872, delivering contextual reasoning and void clause detection (Sec 27, 74) with zero cloud telemetry.
3. **100% Offline Neural Speech (Sherpa-ONNX):** Embedded C++ synthesis running Piper VITS locally via ONNX Runtime. Streams 16-bit PCM audio at 22.05 kHz with 5ms soft anti-pop ramping and zero cloud reliance.
4. **Native Multilingual Support (English, Hindi, Kannada):** Complete explanations, statutory breakdowns, and spoken readouts in English, Hindi (हिंदी), and Kannada (ಕನ್ನಡ) with natural currency pronunciation (₹ → रुपये / ರೂಪಾಯಿ).
5. **Ready-to-Send Counter-Clauses:** Moves beyond passive alerts by drafting exact, legally balanced replacement clauses that you can copy and paste directly into negotiation emails.
6. **Completely Free Forever with Zero Subscriptions:** Runs entirely on your smartphone's processor. Eliminates costly cloud server bills, making contract self-defense permanently free for everyone.

7. Architectural Alignment: Where Edge AI Defends the User



8. Conclusion: The Sovereign Signature

A human signature is an exercise of personal sovereignty. It should represent informed consent, not coerced surrender. When artificial intelligence runs strictly on-device, it answers solely to the human holding the phone—uncompromised by corporate monetization of private data, unhindered by cloud downtime, and untainted by subscription paywalls. By uniting statutory truth, deep human empathy, and uncompromising privacy, VerdictEdge reclaims the sanctity of the human signature. No graduate should sign away their future in panic; no farmer should sign away their land in confusion; and no worker should be cowed by an unenforceable clause. Legal parity is placed where it has always belonged: directly in the hands of the individual.

PROJECT TEAM & ENGINEERING CREDITS

Sai Skanda (Team Lead): Initial architecture, core Android platform & on-device Gemma 2B LLM.

Syed Afnan: Voice transcription pipeline & Google ML Kit on-device OCR integration.

Shreesha K S: OCR pipeline bug fixes, stability optimizations & Sherpa-ONNX local neural TTS engine.

Samarth Sajjan: UI fixes, Jetpack Compose design system & frontend usability polish.

VerdictEdge System Specification & Engineering Manifesto

Core Runtime: Native Android (Kotlin, Jetpack Compose, Material 3, Coroutines)

LLM Inference: Google Gemma 2B INT4 (`gemma-2b-it-gpu-int4.bin` via LiteRT / MediaPipe)

OCR & Vision: Google ML Kit Devanagari OCR & Android PdfRenderer (300 DPI)

Neural Speech: Sherpa-ONNX & Piper VITS (C++ JNI, 16-bit PCM @ 22.05 kHz)

Legal Engine: Indian Contract Act, 1872 (Deterministic Sec 27, 28, 74)

Security & Privacy: Air-Gapped (0 KB Telemetry; `android.permission.INTERNET` omitted)